

GROUP NAME: MOREMI

COHORT: 10

CHALLENGE: 2

NAME OF ACTIVE MEMBERS:

- Adegboyega Babatunde
- Ageebee Thomas
- Elufisan Iyiola
- Faruq Olasupo
- Salim Alhassan
- Abubakar Usman
- Favour Igwezeke
- Osumune Chiamaka
- Samuel Olorunsogbon

DATA CARD

Input vs. Output	User Inputs: Free-text clinical queries, patient symptom descriptions, or anonymised medical case histories (Text). Model Outputs: Plain-language, evidence-based clinical guidance summaries paired with direct citations to peer-reviewed literature, alongside confidence ratings or explicit "low-confidence / refer to physician" flags.	
Classification Problem	Classification Task: Query Intent Classification and Urgency/Triage Categorisation. Categories / Classes: Intent: Symptom Interpretation, Treatment Protocol Summary, Literature Synthesis, or General Health Inquiry. Triage Severity: Emergency/Immediate Care, Urgent Clinical Follow-up, or Routine/Informational.	

	Rationale: Sorting inputs ensures high-risk emergency queries trigger hard safeguards or immediate referral disclaimers rather than generative synthesis.	
Labeling	Label Sources: Domain experts (board-certified clinicians, medical librarians, and clinical researchers). Consistency & Accuracy Verification: Dual-blind review protocols where two independent clinical experts evaluate instruction-tuning datasets. Cohen's Kappa coefficient will be calculated to measure inter-annotator agreement, with a third senior medical specialist resolving discrepancies.	
Data Creation	Collection Method: Sourcing open-access, peer-reviewed medical journals (e.g., PubMed Central, open-access clinical trial repositories) and curated open clinical benchmarks (e.g., PubMedQA, MedQA). Low-Resource / Synthetic Strategy: Synthetic patient query generation created using privacy-preserving LLM prompts grounded strictly in clinical guidelines, followed by expert human validation to expand low-resource symptom phrasing.	
Web Collection Ethics	Web Sourcing: Yes, restricted to public medical literature APIs and open-access databases (e.g., NCBI PubMed API, PMC Open Access Subset). Tools: Automated scraper scripts adhering to robots.txt and API rate limits, combined with human expert filtering.	

	Consent, Copyright, & Privacy: Strictly collecting open-access literature under Creative Commons or public domain licenses. No unverified patient forum scraping is permitted. All training text undergoes automated HIPAA Safe Harbour de-identification pipelines to scrub PII/PHI.	
Bias and Representation	Underrepresented Groups: Patients from low-resource settings, non-native English speakers, and populations with rare or localised symptom expressions. Mitigation Strategy: Active curation of global health datasets (e.g., WHO guidelines) and inclusion of varied clinical dialectal variations in instruction-tuning sets to avoid over-indexing on high-resource Western hospital populations.	
Values and Community Alignment	Value Reflection (Safety, Privacy, Transparency): Privacy is enforced via strict de-identification; Safety is preserved through clinical human-in-the-loop validation; Transparency is embedded by forcing the SLM to output direct citations for all generated claims. Community Benefit & Accountability: Provides clinicians with rapid, reliable reference tools and gives patients safer, non-hallucinatory guidance. Accountability is maintained via public release of the Data Card, regular audit logs, and an open feedback channel for practising clinicians to report inaccuracies.	